

Progress Report

- Increment 2 -

Group #13

Please use this template to describe your progress on the group project in the latest increment. Please do not change the font, font size, margins or line spacing. All the text in italic should be removed from your final submission.

1) Team Members

Owen Proth - oop23 - oproth32

Hubert Pilichowski - hp22q - hpilichowski1

Jasper Darnell - jd21p - jasperdarnell

2) Project Title and Description

The goal of our project is to develop a full-stack application that promotes political and civic engagement at the local level. We recognize that many people either lack understanding of their local government or feel disconnected from it. To address this, our app will make engagement simple, accessible, and interactive, so users don't feel burdened or overwhelmed. It will provide quick, easy access to the most important local issues and opportunities, while also guiding users on concrete steps they can take to start making a difference in their communities.

3) Accomplishments and overall project status during this increment

During this phase of the project, our primary goal was to develop all the foundational components necessary to begin implementing the app's core features. While our team has substantial experience with C++ and other C-based languages, building a full-stack application with both a front-end interface and an integrated database was a new challenge. Because of this, we focused on establishing a strong, well-structured foundation before diving into the feature development phase.

We have nearly completed the front-end design and fully developed the database along with its associated class structures. With these core elements in place, we are now positioned to begin connecting our APIs and implementing functional features using C#. This solid foundation provides us with a clear, tangible framework to guide our next stage of development and ensures smoother integration between components moving forward.

4) Challenges, changes in the plan and scope of the project and things that went wrong during this increment

One of the first challenges we encountered was managing merges across our GitHub pushes. While each of us had prior experience using GitHub individually or for short-term projects, this was our first time collaborating on a large-scale, long-term repository. It took some time to establish an efficient workflow for merging changes, reviewing pull requests, and ensuring no team member's work was overwritten or lost. After some initial trial and error, we developed a process that works smoothly and supports consistent collaboration moving forward.

Another major challenge involved designing the database structure. Although we had previous experience working with databases on a smaller scale, we wanted to ensure our schema could fully support the range of features we plan to implement later. This required significant discussion and research to determine the most efficient and scalable design. Ultimately, this challenge proved valuable — the resulting database architecture is robust, flexible, and well-suited for our future development goals.

We have not made any changes to our original project scope or plan. The structure and framework established early on have continued to meet our needs, reflecting the benefits of the thoughtful planning and collaboration that went into our initial setup.

Toward the end of this increment, we began creating templates for upcoming features and started implementing our first functional component — the login page. During this process, we encountered an issue with writing passwords to the database. Our team is currently working together to resolve this, and because feature development was originally planned for the next increment, we remain ahead of schedule. Once the login page is fully functional, it will serve as a model for integrating the remaining features, turning this obstacle into another learning opportunity for our team.

5) Team Member Contribution for this increment

a) This document was a strong team effort, with each member contributing to a shared understanding of our progress and next steps. Jasper focused primarily on the front-end development, while Owen and Hubert concentrated on the database and setting up the foundational structure for future features. Because of this division of focus, we spent considerable time communicating about our challenges, successes, and overall progress to ensure a well-rounded perspective of where the project currently stands. As we move into the final stages, we continue to discuss our collective vision for the project's direction and upcoming goals. Our aim is to remain a cohesive and collaborative team throughout the entire process, and we plan to meet in person more frequently during this next increment to stay aligned and maintain momentum as we work toward delivering the final product..

b) This stayed mostly the same with only a minor tweak that Owen and Hubert made after working with the database. Jasper looked over it after it was finished to ensure it wasn't missing anything that he added during the increment.

c) This again was a big collaboration. Jasper was in charge of the front end, while Hubert and Owen focused on the database. Therefore everyone had to add what technologies they added. We went through all the requirements together and talked about them all as a team before writing

anything because we all wanted a full understanding of where the project was currently at so that we can move forward as a team in this next increment without any potential blindspots.

d)

Jasper Darnell focused primarily on the front-end application structure in .NET MAUI, implementing the MVVM pattern, page layouts, and the supporting models/services needed to display data in the app. His work is what made the app actually “feel” like a real civic engagement app — with a home dashboard, representatives page, and policies page that all share the same visual style.

Specifically, Jasper:

- *Built the MAUI views (XAML pages) such as the `HomePage.xaml`, `PoliciesPage.xaml`, and `RepresentativesPage.xaml`, giving the application a consistent, modern layout with headers, sections, cards, and scrollable content.*
- *Implemented the corresponding ViewModels (e.g., `HomeViewModel`, `PoliciesViewModel`, `RepresentativesViewModel`) using the MVVM approach so that UI elements were data-bound instead of hardcoded. This makes it easy for the rest of the team to hook in real API data later.*
- *Created reusable models like `CivicStat`, `Policy`, `Representative`, and `VoteChoice` to standardize how civic data is represented across pages.*
- *Set up service interfaces (`IPolicyService`, `IRepresentativeService`) and mock implementations so the UI could be developed and tested even before the backend was fully wired up. This let the team see actual data on screen while Owen finished the API layer.*
- *Added styling resources (`Colors.xaml`, `Styles.xaml`) to centralize colors, typography, and card styles so that every page in the app has the same look and feel.*
- *Implemented UI behaviors such as collection views for “Recent Votes,” civic stats cards, and quick-action buttons, making the home screen a real “dashboard” instead of just a placeholder.*
- *Used data binding and converters (such as the `ConsistencyScoreToColorConverter`) to change UI appearance based on data — for example, showing a different color when a score is high vs. low.*
- *Built similar page structures for multiple sections of the app (as shown in the screenshots), applying the same pattern of: `Model` → `Service` → `ViewModel` → `View`, so the app is scalable as more features (news, quizzes, alignment, etc.) are added.*

- *Collaborated with Hubert on front-end issues (styling, layout, and binding) to make sure the login/register flow and the dashboard pages felt like part of the same application.*

Hubert Pilichowski served as the lead on the data layer and was a key collaborator across both the backend and front-end portions of the system. His primary contribution was researching, designing, and implementing the project's MySQL database schema to support the long-term feature set of the application.

Specifically, Hubert:

- Designed and wrote the full MySQL schema for the application, including core tables such as users, representatives, candidates, legislation, vote_records, and the tables needed for vote simulations, quizzes, news, and the CivicHub/dashboard features. The schema includes proper primary keys, foreign keys, indexing, and cascading rules to keep data consistent and to support later features like comparing user votes to representatives and storing curated news.
- Set up database user and permissions (e.g., creating the politicalapp MySQL user and granting the necessary privileges) so the API and app could connect securely to the database instance.
- Structured the database to match the feature roadmap, not just the current increment — for example, adding quiz/question tables, alignment data, and curated news tables so the team would not have to redesign the database later.
- Implemented and formatted front-end XAML for the authentication flow (login/register tabbed page) in the MAUI app, creating a polished AuthPage with separate tabs for Login and Register, input fields for email/password, password toggle buttons, “remember me,” and placeholders for social login. This gave the team a working UI to hook into Owen's API endpoints.
- Worked closely with Jasper on front-end layout issues, helping resolve binding, layout, and styling questions so the UI stayed consistent across pages.
- Coordinated with Owen on the API side to make sure the database tables, column names, and relationships in MySQL matched what the Entity Framework Core models expected, especially around users, reps, and authentication. This made it easier for the backend to consume the database without major refactors.
- Acted as the main researcher for database best practices in this project, helping the team decide on data types (e.g., CHAR(36) for GUIDs), when to use JSON columns, and how to structure many-to-many relationships (e.g., quizzes and quiz questions).

Owen Proth was primarily responsible for implementing the backend API layer and ensuring seamless integration between the .NET 8 Web API, Entity Framework Core, and the existing MySQL database. His work established the foundation for data communication between the front-end MAUI application and the backend services.

Specifically, Owen:

- Configured the backend startup environment (Program.cs), setting up database connections using the Pomelo MySQL provider, registering the Entity Framework Core context through dependency injection, and enabling development tools such as Swagger for API documentation and testing.
- Developed the AuthController, which manages user registration and login functionality. This included creating secure password handling using BCrypt hashing and verification, implementing validation for duplicate user accounts, and returning structured responses through custom DTOs.
- Implemented the authentication logic that connects front-end user input to backend database actions, enabling reliable user account creation, credential verification, and error handling.
- Built and refined core model classes (such as Candidate, User, and related entities) to ensure consistency with the database schema created by Hubert, supporting proper data relationships within Entity Framework Core.
- Integrated error logging, API routing, and HTTPS middleware, establishing a scalable and maintainable framework for future endpoints and additional feature integration.

e) Jasper provided again a run through of the current front end that we have, Hubert explained the database we implemented, and Owen talked about the backend we implemented plus future goals

6) Plans for the next increment

The next increment will serve as the final phase of development, where our primary focus will be transforming the strong foundation we have built into a fully functional, feature-complete application. Having already established a stable front-end, database, and backend API structure, our goal is now to integrate all components into a seamless, working system.

During this phase, we will begin connecting the front end to the database through our API layer, ensuring smooth data flow between user interactions and stored information. The team will divide responsibilities to efficiently complete all remaining features:

- *Jasper will focus on implementing and polishing the front-end user experience, expanding existing MAUI pages such as the Representative Directory, Policy Summaries, and Civic Hub dashboard. He will ensure consistent styling, navigation, and UI responsiveness across all platforms.*
- *Hubert will lead database integration and optimization, finalizing queries and ensuring that all data from user accounts, representatives, candidates, and simulated votes are correctly stored and retrieved through Entity Framework Core and MySQL.*
- *Owen will focus on external API connections and AI-based functionality, integrating services such as the Google Civic Information API, OpenStates API, BallotReady API, and news sources, as well as implementing AI-generated policy summaries using OpenAI or Hugging Face models.*

Our main deliverables for this increment include the following key features:

- ***Representative Directory:*** *allows users to browse verified profiles, review voting records, and view issue stances.*
- ***Policy Summaries and Voting Simulations:*** *users can read AI-generated summaries of current bills and cast simulated votes to compare with actual representative outcomes.*
- ***Representative Consistency Score:*** *quantifies how closely each representative's actions align with their stated policy positions.*
- ***Belief Alignment Quiz:*** *matches users with representatives and candidates based on shared policy perspectives.*
- ***Personalized Civic Hub:*** *compiles user quiz results, simulated votes, and curated political news into one interactive dashboard.*

The team is very enthusiastic about this next step, as it marks the point where the project evolves from a framework into a fully functional civic engagement platform. By the end of this increment, we aim to deliver a system that not only meets all planned requirements but also demonstrates real potential to help users become more informed and active participants in their local and national communities.

7) Stakeholder Communication

Subject: Progress Update

Dear Stakeholders,

We're pleased to share an update on our progress toward building a digital platform designed to make civic participation more informed, accessible, and engaging. Our application aims to connect citizens with their representatives, simplify legislative tracking, and empower users to explore how their beliefs align with those shaping policy.

Over the past development phase, our team has focused on establishing a solid foundation for the system, one that ensures stability, scalability, and long-term growth. The front-end interface has been largely completed, offering a modern, intuitive design that makes navigation simple for users at all levels of familiarity with politics. Behind the scenes, our database architecture is fully operational, designed to securely store representative profiles, voting records, user accounts, quizzes, and curated news feeds. We have also finalized the API and backend connections necessary to link the app's interface with live data sources.

Like many large-scale civic technology projects, we faced common challenges early on, including merging multiple development branches and refining database relationships to accommodate real-world legislative data. However, these challenges proved invaluable. They allowed us to improve our workflow, strengthen data reliability, and ensure our architecture could handle future integrations with legislative and news APIs.

As we move into the final development phase, our goal is to bring the entire system to life by integrating the user interface, backend, and external data services.

Once these components are unified, users will be able to browse verified representatives, simulate votes on upcoming legislation, track consistency scores, take alignment quizzes, and view personalized civic dashboards, all in one connected platform.

We are confident that this next stage will transform the project from a framework into a fully functional civic engagement application capable of helping citizens engage more directly and meaningfully with their government. We look forward to sharing the finished product and its potential to strengthen the link between policy and participation.

Warm regards,

Group 13 Development Team

Owen Proth • Hubert Pilichowski • Jasper Darnell

8) Link to video

Paste here the link to your video.